

Geometric Nuclear Reaction Theory in CPTG

Deuterium-Proton Capture and Primordial Mass-Seven Transport

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Abstract

This paper extends the Curvature Polarization Transport Gravity (CPTG) lithium solution into a geometric nuclear-reaction framework for deuterium-proton radiative capture. The reaction-specific forward channel is ${}^2\text{H} + p \rightarrow {}^3\text{He} + \gamma$, with the corresponding reverse photodisintegration channel ${}^3\text{He} + \gamma \rightarrow p + {}^2\text{H}$. The construction carries a native geometric source state through a coherent reaction amplitude, a closed-form dimensional astrophysical S -factor branch, a thermonuclear rate interface, and a transported Big Bang nucleosynthesis abundance coordinate. The abundance layer then applies the locked CPTG mass-seven transport rule introduced in the lithium paper: a live source-network gate on ${}^7\text{Li}$ and ${}^7\text{Be}$, with optical depth $\tau_7 = \ln \pi$ and survival factor $1/\pi$. The resulting chain preserves the successful deuterium and helium controls while moving the surviving lithium abundance into the band inferred from old metal-poor stars. The distinct comparison rows are small-residual tests of different layers of the construction: the closed-form $S_{12}(0)$ row lies -0.149960σ from the LUNA-extrapolated comparison value reported by Solar Fusion II, the PRyMordial closed-form locked-gate lithium row gives a $+0.90593\sigma$ residual, and the AlterBBN high-precision rate-response marker, after applying the same fixed mass-seven gate for comparison display, gives a lithium residual of $+1.15975\sigma$. The AlterBBN row is not used as an independent full abundance-admission row. These rows are not competing values for a single calculation. They are separate checks in the source-to-reaction-to-abundance evidence chain. The result is a reaction-specific geometric nuclear theory and abundance-transport extension of the CPTG lithium solution, not a claim that all nuclear reactions are solved without separate source construction.

1 Introduction

This paper is an extension of the CPTG paper titled “The Cosmological Lithium Problem and the CPTG Lithium Solution” [11]. That earlier work addressed the observed mass-seven discrepancy as a transport problem in the primordial abundance network. The present paper asks a broader question: whether the same geometric framework can also be carried through a nuclear reaction channel, from a native source state to a reaction amplitude, an astrophysical S -factor, a thermonuclear rate interface, and an abundance comparison.

The selected commissioning reaction is deuterium-proton radiative capture,



The corresponding reverse channel is



This reaction is a useful test case because it sits at the intersection of few-body nuclear structure, charged-particle capture, electromagnetic emission, solar-fusion comparison, and big-bang nucleosynthesis. It is therefore simple enough to make the formulas explicit, but important enough that a successful geometric construction has physical meaning.

The guiding idea is straightforward. CPTG treats curvature response and curvature transport as geometric structure rather than as a collection of unrelated fitted corrections. In the nuclear setting, the question becomes whether a reaction-specific source state can be organized geometrically and then carried into the observables used by nuclear astrophysics. The observables are conventional: an astrophysical S -factor, a thermonuclear rate, deuterium and helium abundance controls, helium-three consistency, and the lithium abundance inferred from old metal-poor stars.

This point is important because the paper is not only a lithium-abundance note. The lithium solution supplies the locked live $A = 7$ transport rule, but the new content is the geometric nuclear reaction chain. The first part explains why surviving mass seven is transported by a fixed optical depth. The second part supplies a native source-to-amplitude construction for $D(p, \gamma){}^3\text{He}$. The third part tests the resulting framework against nuclear and abundance observables.

The significance of the present construction is not merely that it proposes a geometric correction or that it reduces lithium. The benchmark is whether a single framework can connect a stated nuclear source-state construction to a coherent $D(p, \gamma){}^3\text{He}$ reaction amplitude, a closed-form S -factor comparison, a BBN network interface that preserves D/H and Y_p , and a fixed live $A = 7$ transport rule that moves the raw mass-seven excess into the lithium comparison band inferred from old metal-poor stars. Many existing lithium-problem solution classes are formulated primarily at the abundance, stellar-depletion, expansion-history, or nonstandard-particle level. The comparison made here is narrower and more structural: whether one fixed construction can also supply a stated nuclear source-state path, a coherent reaction amplitude, an S -factor contact row, and a live mass-seven transport rule. This makes the framework explicit, falsifiable, and resistant to being reduced to a single lithium-tuning parameter.

This result is strengthened by the comparison-coordinate discipline inherited from the broader CPTG program [9, 10]. The geometric branch is fixed first, then projected into the observational coordinate. The present paper therefore does not introduce a new lithium-only parameter. It asks whether the same locked geometric branch can survive a nuclear source-state construction, an S -factor comparison, a transported BBN coordinate, and a live $A = 7$ source-network gate without sector-by-sector retuning.

Layer	Fixed input	Derived output	External test	Retuning allowed?
CPTG branch	$p_C, p_{ac}, G_T, 1/\pi$	Transported BBN coordinate	D/H and Y_p controls	No
$A = 3$ source state	68 support indices and four scalar components	$\mathcal{A}_{C6}, I_{C7}$	Source and amplitude audit	No
Nuclear comparison	Closed-form CPTG dimensional branch	$S_{12}^{\text{CPTG}}(0)$ row	LUNA comparison row	No
Network layer	Transported BBN coordinate	Raw high mass-seven output	Lithium problem reproduced before gate	No
Mass-seven gate	$\tau_7 = \ln \pi$	$Y_{7,\text{raw}}/\pi$	Lithium abundance band	No

Table 1: Evidence-chain and no-retuning map for the commissioned reaction and abundance calculation. The rows separate fixed inputs, derived quantities, comparison tests, and allowed movement.

The standard BBN picture already succeeds for deuterium and helium, so any proposed solution to the lithium problem must be more constrained than a simple reduction of lithium.

The dimensional S_{12} bridge is the closed-form CPTG normalization $K_{S0} = \alpha\pi/(33 - 1/(7\pi)) = 0.0006956649414530029 \text{ eV b}$, using the companion-parameter-ledger convention $\alpha^{-1} = 137.035999177$. It gives $S_{12}^{\text{CPTG}}(0) = 0.2145003969219307 \text{ eV b}$ and $S_{12}^{\text{CPTG}}(6.64 \text{ keV}) = 0.2516322073981925 \text{ eV b}$. That value is the operative paper value throughout. It keeps D/H and Y_p in their control bands in the transported PRyMordial row, creates no free lithium-depletion parameter, and preserves the measured nuclear capture comparison.

The CPTG construction satisfies these requirements in the presented comparison rows: the closed-form $S_{12}(0)$ comparison row is -0.149960σ , the PRyMordial locked-gate lithium row is $+0.90593\sigma$, and the AlterBBN high-precision marker row resolves this same branch, $K_{S0} = 0.0006956649414530029 \text{ eV b}$ with $S_{12}^{\text{CPTG}}(0) = 0.2145003969219307 \text{ eV b}$, in a second network implementation. These are small signed pulls and computed high-precision network responses, not marginal high-sigma survivals.

2 Native Geometric Ordering

CPTG separates native geometric quantities from observational comparison coordinates. The nuclear reaction chain used here follows the hierarchy

$$\begin{array}{c}
 \text{CPTG native geometry} \\
 \downarrow \\
 \text{reaction source amplitude} \\
 \downarrow \\
 S(E) \\
 \downarrow \\
 N_A \langle \sigma v \rangle \\
 \downarrow \\
 \text{abundance comparison.}
 \end{array} \tag{3}$$

This ordering prevents the abundance comparison from being mistaken for the native theory. The geometric source state supplies the reaction amplitude. The astrophysical S -factor introduces a dimensional comparison layer. The thermonuclear rate places the channel into a BBN network. The abundance layer then tests deuterium, helium, helium-three, and mass-seven outputs against observational comparison values.

For the mass-seven sector, the native abundance ordering is

$$\begin{array}{c}
 \text{curvature transport} \\
 \downarrow \\
 \text{transported BBN coordinate} \\
 \downarrow \\
 \text{live } A = 7 \text{ source gate} \\
 \downarrow \\
 \text{observed light elements.}
 \end{array} \tag{4}$$

The live gate is not an external post-processing stellar depletion factor and not a division applied to the final printed abundance after the network has finished. The live ${}^7\text{Li}$ and ${}^7\text{Be}$ variables are evolved with the transport term inside the source-network layer. The raw mass-seven excess is therefore still visible before the gate, and the final lithium comparison tests whether the fixed geometric optical depth carries the live $A = 7$ channels into the abundance band inferred from old metal-poor stars while D/H and Y_p remain admitted controls. In this paper, the old metal-poor star comparison enters through the observational lithium band.

The $D(p, \gamma){}^3\text{He}$ construction and the $A = 7$ transport gate play different roles. The first tests whether CPTG can be carried from a nuclear source-state construction into a reaction-strength and rate layer. The second tests whether the same geometric transport framework can act on the surviving mass-seven abundance channels without disturbing the admitted D/H and Y_p background. The paper therefore does not use lithium to define

the $D(p, \gamma)^3\text{He}$ amplitude, and it does not use the S -factor normalization as a lithium depletion parameter.

3 Reaction Definition and Charged-Particle Kinematics

For a charged two-body entrance channel, the cross section is conventionally written in terms of the astrophysical S -factor as

$$\sigma(E) = \frac{S(E)}{E} \exp[-2\pi\eta(E)]. \quad (5)$$

Here E is the center-of-mass energy and $\eta(E)$ is the Sommerfeld parameter. For charges Z_1 and Z_2 , reduced mass μ , and fine-structure constant α ,

$$\eta(E) = Z_1 Z_2 \alpha \sqrt{\frac{\mu c^2}{2E}}. \quad (6)$$

For $D(p, \gamma)^3\text{He}$, the entrance-channel charges are

$$Z_1 = 1, \quad Z_2 = 1. \quad (7)$$

The reduced mass is

$$\mu = \frac{m_p m_D}{m_p + m_D}. \quad (8)$$

The thermonuclear rate is then obtained from the Maxwellian average

$$N_A \langle \sigma v \rangle = N_A \left(\frac{8}{\pi \mu} \right)^{1/2} \frac{1}{(k_B T)^{3/2}} \int_0^\infty S(E) \exp \left[-\frac{E}{k_B T} - 2\pi\eta(E) \right] dE. \quad (9)$$

This expression defines the bridge from the source-normalized $S(E)$ channel into a reaction-network rate. Any tabulated network implementation must preserve the same physical interpretation of the charged-particle entrance channel, the reverse detailed-balance relation, and the abundance variables to which the rate couples.

4 Fixed Comparison Parameters and Symbols

The cosmology-facing abundance coordinate uses the geometric pi-branch transport map. The native curvature exponent is

$$p_C = \pi. \quad (10)$$

The acoustic comparison exponent is

$$p_{\text{ac}} = 3 - \frac{\pi}{100} = 2.968584073464. \quad (11)$$

The acoustic transport factor is

$$G_T = \frac{p_{ac}}{p_C} = 0.9449296585513721. \quad (12)$$

The square-root transport factor is

$$\sqrt{G_T} = 0.9720749243506758. \quad (13)$$

The transported BBN abundance coordinate is

$$\eta_{10}^{\text{BBN}} = \eta_{10}^{\text{ac}} \sqrt{G_T}. \quad (14)$$

Numerically,

$$\eta_{10}^{\text{BBN}} = 5.998071834744. \quad (15)$$

The corresponding physical baryon-density coordinate is

$$\Omega_b h_{\text{BBN}}^2 = 0.021898765370. \quad (16)$$

Symbol	Meaning	Value or definition	Role
p_C	Native curvature exponent	π	Native branch
p_{ac}	Acoustic comparison exponent	$3 - \pi/100 = 2.968584073464$	Acoustic layer
G_T	Transport factor	$p_{ac}/p_C = 0.9449296585513721$	BBN and acoustic transport
$\sqrt{G_T}$	Square-root transport factor	0.9720749243506758	BBN coordinate conversion
η_{10}^{BBN}	Transported baryon-to-photon coordinate	5.998071834744	BBN abundance coordinate
$\Omega_b h_{\text{BBN}}^2$	Transported physical baryon-density coordinate	0.021898765370	BBN abundance coordinate
τ_7	Mass-seven transport optical depth	$\ln \pi$	Locked live $A = 7$ gate
f_7	Mass-seven survival factor	$1/\pi = 0.3183098861837907$	Lithium comparison

Table 2: Core symbols and fixed comparison parameters used in the reaction and abundance layers.

Quantity	Classification	Meaning in the paper
$p_C = \pi$	fixed branch quantity	Native curvature exponent of the comparison branch.
$G_T = p_{ac}/p_C$	fixed transport conversion	Converts the acoustic comparison layer into the transported BBN coordinate.
η_{10}^{BBN}	transported coordinate	BBN abundance coordinate; not a lithium depletion factor.
$\mathcal{A}_{C6}$	derived source output	Coherent source amplitude obtained from the fixed source state and coefficient sector.
I_{C7}	derived intensity output	Dimensionless source intensity before dimensional S-factor normalization.
K_{S0}	closed-form conversion	Dimensional conversion from source intensity to the zero-energy S-factor scale.
$S_{\text{CPTG}}(0)$	closed-form comparison	Zero-energy S-factor used for the $D(p, \gamma)^3\text{He}$ comparison.
$\tau_7 = \ln \pi$	locked transport rule	Integrated optical depth of the live mass-seven gate.
$Y_{7,\text{CPTG}} = Y_{7,\text{raw}}/\pi$	locked survival rule	Final mass-seven abundance after the fixed live source-network gate.
D/H, Y_p , and ${}^7\text{Li}/H$ pulls	observational tests	External abundance comparisons that can accept or reject the construction.

Table 3: Classification of fixed, closed-form-normalized, derived, and tested quantities. The table separates native branch inputs, source-derived outputs, closed-form dimensional comparisons, and observational tests.

5 Native $A=3$ Source-State Construction

The commissioned reaction channel uses a table-certified $A = 3$ bound-state support with 68 native support indices. Four scalar source components are retained at each support index, giving

$$N_{\text{slot}} = 68 \times 4 = 272. \quad (17)$$

The support-state scalar tuple is written using the signed table energy,

$$\mathcal{S}_i = (T_{A=3}, V_{\text{NN}}, V_{3\text{NF}}, E_{A=3})_i. \quad (18)$$

When a positive binding scale is required in the contraction, the manuscript uses $|E_{A=3}| = 7.73945407385108 \text{ MeV}$. Thus $E_{A=3}$ denotes the signed table energy, while $|E_{A=3}|$ denotes the positive binding-energy scale used for normalization.

The retained scalar values are

$$T_{A=3} = 27.0897 \text{ MeV}, \quad (19)$$

$$V_{\text{NN}} = -34.8757 \text{ MeV}, \quad (20)$$

$$V_{3\text{NF}} = 0.0465074 \text{ MeV}, \quad (21)$$

$$E_{A=3} = -7.73945407385108 \text{ MeV}. \quad (22)$$

The direct energy sum is

$$T_{A=3} + V_{\text{NN}} + V_{\text{3NF}} = -7.7394926000000001 \text{ MeV}. \quad (23)$$

The energy-balance residual is

$$\Delta E = |(T_{A=3} + V_{\text{NN}} + V_{\text{3NF}}) - E_{A=3}| = 3.8526148920858816 \times 10^{-5} \text{ MeV}. \quad (24)$$

Quantity	Value	Interpretation
$T_{A=3}$	27.0897 MeV	Kinetic contribution in the retained $A = 3$ source state
V_{NN}	-34.8757 MeV	Native two-body nuclear contribution
V_{3NF}	0.0465074 MeV	Three-nucleon-force contribution
$E_{A=3}$	-7.73945407385108 MeV	Signed bound-state energy comparison value
ΔE	$3.8526148920858816 \times 10^{-5}$ MeV	Energy-balance residual for the retained state

Table 4: Native $A = 3$ source-state scalar values used to form the geometric reaction source.

The retained row is the table-certified NUCWF row-362 state identified in the source-state evidence ledger. Its row hashname is printed in Table 5, with $\text{NORM} = 0.99993$, $\text{NP3} = 68$, 272 support-slot scalar records, and 340 coherent amplitude terms. The energy residual printed above is the display-precision manuscript reconstruction from the displayed scalar values; the full reproducibility route is the companion report archive named in Section 18.

Ledger item	Retained value	Role
NUCWF row index	362	Table-certified $A = 3$ source state
Row hashname	4faa3edcfbff73257c73efade1c5e1d acab105fad4148eecb3d982919213a5e6	Row identity in the source-state evidence ledger
State norm	0.99993	Retained normalization check
Structural-zero floor	0.00010005000000500001 MeV	Row-specific energy-residual admission floor
Energy residual	0.000038526148920858816 MeV	Below the row-specific floor

Table 5: Retained row-362 source-state identity and audit values recovered from the source-state evidence ledger.

6 Current and Source Coefficients

The reaction source uses five retained current/source coefficients,

$$\mathbf{c} = (c_1, c_3, c_4, c_D, c_E). \quad (25)$$

Numerically,

$$\mathbf{c} = (-1.23, -4.65, 3.28, 0.8918, -0.38595). \quad (26)$$

The coefficients c_1 , c_3 , and c_4 carry units of GeV^{-1} . The coefficients c_D and c_E are dimensionless. They enter the source-amplitude construction and are not adjusted to fit the final lithium abundance. The numerical tuple corresponds to the retained $N4LO+$, $\Lambda_N = 450 \text{ MeV}$ source-coefficient set used in the supporting $A = 3$ construction; the long-range c_i carry GeV^{-1} units and c_D , c_E are dimensionless [8].

Coefficient	Value	Unit	Role
c_1	-1.23	GeV^{-1}	Long-range current/source coefficient
c_3	-4.65	GeV^{-1}	Long-range current/source coefficient
c_4	3.28	GeV^{-1}	Long-range current/source coefficient
c_D	0.8918	dimensionless	Short-range current/source coefficient
c_E	-0.38595	dimensionless	Short-range source coefficient

Table 6: Current/source coefficients used in the coherent reaction-amplitude construction.

With 68 support indices and five coefficients, the coherent amplitude contains

$$N_{\text{amp}} = 68 \times 5 = 340 \quad (27)$$

source-weighted terms.

7 Coherent Amplitude and Source Intensity

The reduced coherent pre-rate amplitude is written as

$$\mathcal{A}_{C6} = \sum_{i=1}^{68} \sum_{k=1}^5 W_{ik} c_k \mathcal{S}_{ik}. \quad (28)$$

Here W_{ik} is the fixed dimensionless source-weight map for support index i and coefficient channel k . It is not a fitted reaction coefficient. It carries the support-grid projection, six-map pullback, adjoint transport, operator-support selection, and channel normalization into the scalar source-amplitude contraction,

$$W_{ik} \equiv \left\langle \chi_i^{(A=3)} \left| \mathcal{P}_k^{\text{geom}} \right| \chi_i^{(pD)} \right\rangle_{\text{support}}. \quad (29)$$

The symbol $\chi_i^{(A=3)}$ denotes the retained bound-state support component, $\chi_i^{(pD)}$ denotes the corresponding proton-deuterium support component in the reaction channel, and $\mathcal{P}_k^{\text{geom}}$ denotes the fixed geometric projection for coefficient channel k . Once the source state, support registry, and current/source coefficient set are fixed, W_{ik} is fixed for the commissioned reaction channel. The factor c_k is the current/source coefficient, and $\mathcal{S}_{ik}$ is the scalar source component associated with support index i and coefficient channel k .

The apparent four-scalar/five-coefficient indexing is resolved by a coefficient-channel projection. The four retained row-362 scalars supply the kinetic, two-body, three-body, and

signed-energy information; the fifth coefficient channel is the closure-residual channel. In the retained contraction rule, the coefficient-channel scalars are

$$\mathcal{S}_{i1} = \frac{T_{A=3}}{|E_{A=3}|}, \quad (30)$$

$$\mathcal{S}_{i2} = \frac{V_{\text{NN}}}{|E_{A=3}|}, \quad (31)$$

$$\mathcal{S}_{i3} = \frac{V_{\text{3NF}}}{|E_{A=3}|}, \quad (32)$$

$$\mathcal{S}_{i4} = \frac{|E_{A=3}|}{|E_{A=3}|}, \quad (33)$$

$$\mathcal{S}_{i5} = \frac{T_{A=3} + V_{\text{NN}} + V_{\text{3NF}} + |E_{A=3}|}{|E_{A=3}|}. \quad (34)$$

Equivalently, the source-ledger contraction can be written as

$$\mathcal{A}_{C6} = \sqrt{\text{NORM}} \sum_i \frac{1}{68} \left[c_1 \frac{T}{|E|} + c_3 \frac{V_{\text{NN}}}{|E|} + c_4 \frac{V_{\text{3NF}}}{|E|} + c_D \frac{|E|}{|E|} + c_E \frac{T + V_{\text{NN}} + V_{\text{3NF}} + |E|}{|E|} \right]. \quad (35)$$

Thus $\mathcal{S}_{ik}$ in the amplitude equation is the already-projected coefficient-channel scalar, not an additional fifth independent source-state scalar.

The printed manuscript defines the contraction but does not inline the full finite 68×5 source-weight map. The full W_{ik} map, retained support-index registry, and contraction audit must be supplied with the companion reproducibility archive identified in the data ledger. A reproduction attempt is successful only if it reconstructs the amplitude and derived intensity from that support map before using the closed-form S -factor normalization. The retained coherent value is

$$\mathcal{A}_{C6} = 17.5595746398194. \quad (36)$$

The corresponding dimensionless source intensity is

$$I_{C7} = |\mathcal{A}_{C6}|^2. \quad (37)$$

Numerically,

$$I_{C7} = 308.3386615313886. \quad (38)$$

This intensity is a native dimensionless source quantity. It requires a dimensional comparison layer before it can be stated as an astrophysical S -factor.

8 Astrophysical S-Factor Comparison

The zero-energy comparison is written as

$$S_{\text{CPTG}}(0) = K_{S0} I_{C7}. \quad (39)$$

The operative dimensional normalization is the closed-form CPTG expression

$$K_{S0} = \frac{\alpha\pi}{33 - \frac{1}{7\pi}} \text{ eV b}. \quad (40)$$

The numerical value used throughout this manuscript is evaluated with the companion-parameter-ledger fine-structure convention

$$\alpha^{-1} = 137.035999177, \quad \alpha = 0.0072973525643314250. \quad (41)$$

In the retained support split this same expression is

$$K_{S0} = \frac{\alpha\pi}{3(NP3B - 1) - \frac{1}{(NP3A/8)\pi}} \text{ eV b}, \quad (42)$$

with

$$NP3A = 56, \quad NP3B = 12, \quad NP3 = 68. \quad (43)$$

The denominator is not fitted to the LUNA-extrapolated comparison value reported by Solar Fusion II. It is the retained support-normalization count for the commissioned $A = 3$ branch: the $NP3B = 12$ block contributes the $3(NP3B - 1)$ support-channel count, while the $NP3A = 56$ block contributes the reciprocal $(NP3A/8)\pi$ geometric correction. With $NP3A = 56$ and $NP3B = 12$, this reduces to $33 - 1/(7\pi)$. The comparison to $S_{12}(0)$ is performed only after this closed-form dimensional branch is fixed.

Numerically,

$$K_{S0} = 0.0006956649414530029 \text{ eV b}. \quad (44)$$

The resulting CPTG zero-energy value is

$$S_{\text{CPTG}}(0) = K_{S0} I_{C7} = 0.2145003969219307 \text{ eV b}. \quad (45)$$

Solar Fusion II reports the LUNA-extrapolated zero-energy comparison value [1]:

$$S_{12,\text{LUNA}}(0) = 0.216 \pm 0.010 \text{ eV b}. \quad (46)$$

The signed residual is

$$\Delta_S = \frac{0.2145003969219307 - 0.216}{0.010} = -0.1499603078069277. \quad (47)$$

At the representative solar Gamow energy,

$$E_0 = 6.64 \text{ keV}, \quad (48)$$

the transported comparison value is

$$S_{\text{CPTG}}(E_0) = 0.2516322073981925 \text{ eV b}. \quad (49)$$

This value uses the Solar Fusion II low-energy shape carrier at $E_0 = 6.64 \text{ keV}$, normalized by the closed-form CPTG zero-energy branch:

$$S_{\text{CPTG}}(E_0) = S_{\text{CPTG}}(0) F_{\text{SFII}}(E_0), \quad F_{\text{SFII}}(6.64 \text{ keV}) = 1.1731083532203264. \quad (50)$$

The energy-shape carrier is therefore external comparison-layer shape information; the zero-energy normalization remains the closed-form CPTG value. Solar Fusion II gives the low-energy best-shape carrier as

$$S_{12,\text{SFII}}(E) = 0.2145 + 5.5612 \left(\frac{E}{\text{MeV}} \right) + 4.6581 \left(\frac{E}{\text{MeV}} \right)^2 \text{ eV b}. \quad (51)$$

Thus the factor used above is

$$F_{\text{SFII}}(E_0) = \frac{S_{12,\text{SFII}}(E_0)}{S_{12,\text{SFII}}(0)} = 1.1731083532203264 \quad (52)$$

for $E_0 = 0.00664 \text{ MeV}$.

Solar Fusion II gives the corresponding solar-Gamow comparison band at this energy as

$$S_{12}(E_0 = 6.64 \text{ keV}) = 0.252 \pm 0.018 \text{ eV b}, \quad (53)$$

so the closed-form CPTG value has residual

$$\Delta_{S,E_0} = \frac{0.2516322073981925 - 0.252}{0.018} = -0.020433. \quad (54)$$

For source-network insertion, the closed-form normalization defines the rate branch

$$R_{pD}^{\text{CPTG}}(T) = \lambda_{pD} R_{pD}^{\text{table}}(T), \quad \lambda_{pD} = 1.0000018504518913. \quad (55)$$

The corresponding network carrier normalization is

$$S_{12}^{\text{table}}(0) = 0.2145000000000000 \text{ eV b}, \quad (56)$$

so that

$$\lambda_{pD} = \frac{S_{12}^{\text{CPTG}}(0)}{S_{12}^{\text{table}}(0)} = \frac{0.2145003969219307}{0.2145000000000000} = 1.0000018504518913. \quad (57)$$

Here $R_{pD}^{\text{table}}(T)$ denotes the unmodified tabulated thermal-rate carrier inside the named abundance-network implementation. The factor λ_{pD} is the fixed ratio between the closed-form CPTG $S_{12}(0)$ branch and that carrier normalization. It is not refitted inside the BBN abundance calculation. The operative theory value is $R_{pD}^{\text{CPTG}}(T)$, defined by the closed-form CPTG normalization above.

Observable	CPTG value	Comparison value	Pull or status
K_{S0}	0.0006956649414530029 eV b	$\alpha\pi/(33 - 1/(7\pi))$	closed-form default
$S_{12}(0)$	0.2145003969219307 eV b	0.216 ± 0.010 eV b	-0.149960σ
$S_{12}(E_0)$, $E_0 = 6.64$ keV	0.2516322073981925 eV b	Solar Fusion II: 0.252 ± 0.018 eV b	-0.02043σ , PASS
Rate insertion factor	$\lambda_{pD} = 1.0000018504518913$	$S_{12}^{\text{table}}(0) =$ 0.2145000000000000 eV b	closed-form CPTG rate branch

Table 7: Closed-form S -factor comparison rows for the commissioned reaction channel. All operative values use the closed-form normalization.

8.1 Raw AlterBBN Marker Output Before the Mass-Seven Gate

The closed-form normalization is the operative CPTG rate scale for the $D(p, \gamma)^3\text{He}$ channel. The abundance-network row below records the raw high-precision AlterBBN marker output obtained after inserting the closed-form CPTG rate branch into the source network. This is a rate-propagation diagnostic, not the final fixed-gate lithium comparison. The fixed mass-seven gate is applied later in the network implementation cross-check table.

Observable	Closed-form CPTG network value	Status
Y_p	$2.4728924324539345 \times 10^{-1}$	raw high-precision marker output
D/H	$2.4348223639496304 \times 10^{-5}$	raw high-precision marker output
$^3\text{He}/H$	$1.0311437909300730 \times 10^{-5}$	raw high-precision marker output
$^7\text{Li}/H$	$5.4661777402483630 \times 10^{-10}$	raw high-precision marker output
$^6\text{Li}/H$	$1.0726676158966467 \times 10^{-14}$	raw high-precision marker output
$^7\text{Be}/H$	$5.1798681429209550 \times 10^{-10}$	raw high-precision marker output

Table 8: Raw high-precision AlterBBN marker output produced after insertion of the closed-form CPTG $D(p, \gamma)^3\text{He}$ rate branch. This table reports the pre-gate network state. The final fixed-gate lithium comparison is reported separately in Table 12.

9 BBN Abundance Coordinate

The abundance comparison is evaluated in the transported BBN coordinate. The acoustic baryon coordinate is not directly reused as a BBN abundance coordinate. Instead,

$$\eta_{10}^{\text{BBN}} = 5.998071834744. \quad (58)$$

The companion coordinate ledger gives the acoustic input and transport rule as

$$\eta_{10}^{\text{ac}} = 6.17038016771246, \quad \sqrt{G_T} = 0.9720749243506758, \quad (59)$$

with

$$\eta_{10}^{\text{BBN}} = \eta_{10}^{\text{ac}} \sqrt{G_T}. \quad (60)$$

The baryon-density coordinate is reported using the retained conversion

$$\Omega_b h_{\text{BBN}}^2 = \frac{\eta_{10}^{\text{BBN}}}{273.9}. \quad (61)$$

The corresponding transported baryon density coordinate is

$$\Omega_b h_{\text{BBN}}^2 = 0.021898765370. \quad (62)$$

At this coordinate, using the closed-form $D(p, \gamma)^3\text{He}$ normalization, the retained raw abundance projection before the live $A = 7$ gate is

$$\frac{D}{H} = 2.5074364788625683 \times 10^{-5}, \quad (63)$$

$$\frac{{}^3\text{He}}{H} = 1.0489357072448880 \times 10^{-5}, \quad (64)$$

$$Y_{p,\text{raw}} = 2.4569433563924720 \times 10^{-1}, \quad (65)$$

$$\left(\frac{{}^7\text{Li}}{H}\right)_{\text{raw}} = 5.2668261732457650 \times 10^{-10}. \quad (66)$$

The raw lithium value is high in the same sense as the standard BBN mass-seven excess. The live mass-seven transport gate is introduced only after the background controls are retained.

10 Locked Live Mass-Seven Transport Gate

The surviving mass-seven abundance is

$$Y_7 = Y_{7\text{Li}} + Y_{7\text{Be}}. \quad (67)$$

The live transport gate acts as an optical-depth factor on the surviving mass-seven channels,

$$Y_{7,\text{final}} = Y_{7,\text{raw}} \exp \left[- \int \Gamma_{7,\text{trans}}(t) dt \right]. \quad (68)$$

The CPTG locked geometric condition is

$$\int \Gamma_{7,\text{trans}}(t) dt = \ln \pi. \quad (69)$$

Therefore,

$$Y_{7,\text{CPTG}} = Y_{7,\text{raw}} e^{-\ln \pi} = \frac{Y_{7,\text{raw}}}{\pi}. \quad (70)$$

The survival factor is

$$\frac{1}{\pi} = 0.3183098861837907. \quad (71)$$

At source-network level the lithium-seven channel is evolved as

$$\frac{dY_{7\text{Li}}}{dt} \longrightarrow \left(\frac{dY_{7\text{Li}}}{dt} \right)_{\text{BBN}} - \Gamma_{7,\text{trans}} Y_{7\text{Li}}. \quad (72)$$

The beryllium-seven channel is evolved as

$$\frac{dY_{7\text{Be}}}{dt} \longrightarrow \left(\frac{dY_{7\text{Be}}}{dt} \right)_{\text{BBN}} - \Gamma_{7,\text{trans}} Y_{7\text{Be}}. \quad (73)$$

The thermal shape or support of $\Gamma_{7,\text{trans}}$ may be varied as a robustness control, but the integrated optical depth is fixed by $\ln \pi$. Those variations are diagnostic stress tests only; the accepted comparison row uses the fixed integrated optical depth and does not select a thermal shape after seeing the lithium result. Thus the gate is not a fitted depletion coefficient.

For the retained closed-form abundance projection, the live-gated lithium value is

$$\left(\frac{{}^7\text{Li}}{H} \right)_{\text{gated}} = 1.6764828397556692 \times 10^{-10}. \quad (74)$$

The live-to-raw ratio is

$$\frac{({}^7\text{Li}/H)_{\text{gated}}}{({}^7\text{Li}/H)_{\text{raw}}} = 0.3183098861837907. \quad (75)$$

This ratio is the locked survival factor itself.

11 Observed Comparison Protocol

For any observable X , the signed pull is

$$\Delta_X = \frac{X_{\text{CPTG}} - X_{\text{obs}}}{\sigma_X}. \quad (76)$$

The abundance admission and comparison rule used in this paper is

$$|\Delta_X| < 2. \quad (77)$$

This inequality is only the admission rule. The numerical comparison rows below still report the actual signed pulls, including the retained closed-form abundance projection values -0.01943σ for D/H , $+0.23145\sigma$ for Y_p , $+15.2673\sigma$ for raw lithium, and $+0.90593\sigma$ for gated lithium.

The observational comparison values are the Review of Particle Physics Big-Bang nucleosynthesis recommended values [7, 5, 6]

$$\left(\frac{{}^7\text{Li}}{H}\right)_{\text{obs}} = (1.45 \pm 0.25) \times 10^{-10}, \quad (78)$$

$$\left(\frac{D}{H}\right)_{\text{obs}} = (25.08 \pm 0.29) \times 10^{-6}, \quad (79)$$

$$Y_{p,\text{obs}} = 0.245 \pm 0.003. \quad (80)$$

Helium-three is treated as an advisory diagnostic rather than a symmetric central-value pull in the comparison table. In this manuscript it is not assigned a central observational value with a Gaussian uncertainty, so no signed σ pull is reported for that row. The Review of Particle Physics BBN review cautions that ${}^3\text{He}$ is not presently a clean cosmological-probe abundance in the same sense as D/H or Y_p ; the ${}^3\text{He}/H$ row is therefore reported for diagnostic continuity, not as an independent admission observable [6].

The lithium comparison band is inferred from old metal-poor stars and is used here as the reported comparison quantity for surviving primordial mass seven. The Review of Particle Physics BBN review treats this lithium value as a stellar-inference comparison with a lower-bound caveat rather than as a direct primordial measurement. CPTG is therefore tested against the lithium abundance band inferred from those stars, but the mechanism used in this paper is not a fitted stellar-surface depletion prescription and does not claim to eliminate all stellar-depletion systematics. The mechanism is the fixed live source-network transport rule in the primordial abundance calculation [7].

Observable	CPTG or raw value	Obs comparison value, band, or limit	Pull or status
D/H	$2.5074364788625683 \times 10^{-5}$	$(25.08 \pm 0.29) \times 10^{-6}$	-0.01943σ pass
${}^3\text{He}/H$	$1.0489357072448880 \times 10^{-5}$	advisory diagnostic; no central σ pull	N/A as an admission obs
Y_p	$2.4569433563924720 \times 10^{-1}$	0.245 ± 0.003	$+0.23145\sigma$ pass
raw ${}^7\text{Li}/H$	$5.2668261732457650 \times 10^{-10}$	$(1.45 \pm 0.25) \times 10^{-10}$	$+15.2673\sigma$ excess
gated ${}^7\text{Li}/H$	$1.6764828397556692 \times 10^{-10}$	$(1.45 \pm 0.25) \times 10^{-10}$	$+0.90593\sigma$ pass

Table 9: Raw and gated abundance comparison in the transported BBN coordinate using the closed-form $D(p, \gamma){}^3\text{He}$ normalization. The raw lithium row displays the standard mass-seven excess, while the locked live gate moves the lithium comparison to a $+0.90593\sigma$ residual without spoiling the deuterium or helium controls. Helium-three is retained as an advisory diagnostic rather than as a symmetric central-value pull or admission observable.

12 Controls and Robustness Tests

The theory is not tested by a single final lithium value. It is tested by a layered control sequence. The source state must close, the support map must remain consistent, the

coherent amplitude must form, the closed-form $S_{12}(0)$ comparison must remain compatible with the external comparison row, the transported BBN coordinate must preserve deuterium and helium, the raw network must still display the mass-seven excess, and only then may the locked live $A = 7$ gate be judged against the lithium abundance inferred from old metal-poor stars.

The control sequence is listed explicitly in Table 10, so the failure gates are visible without treating the paper as a single endpoint fit.

Each layer can fail independently. A source-state failure would prevent a reaction-amplitude claim. An S-factor failure would prevent the commissioned reaction channel from making contact with nuclear astrophysics. A D/H or Y_p failure would reject the transported BBN background before lithium is judged. A raw mass-seven value that was already low would weaken the claim that the lithium problem is being addressed rather than hidden. A live-gate failure would show that the fixed $\ln \pi$ optical depth is not sufficient. A one-code-only success would leave the result more vulnerable to implementation-specific behavior.

Control layer	Requirement	Status in this paper
Source state	The $A = 3$ state must admit a closed support construction.	Closed source-state construction.
Support map	The support registry and contraction map must be fixed before comparison.	Fixed support and W_{ik} contraction rule.
Amplitude	The source state and coefficients must form a coherent amplitude.	$\mathcal{A}_{C6} = 17.5595746398194$.
S-factor	The commissioned reaction must contact an external S-factor comparison.	$\Delta_S = -0.149960\sigma$.
BBN background	D/H and Y_p must remain admitted before lithium is judged.	Deuterium and helium controls retained.
Raw mass seven	The ungated network should still display the lithium excess.	Raw ${}^7\text{Li}/H$ remains high.
Live gate	The fixed $\ln \pi$ gate must move mass seven into the observed band.	Gated lithium passes with actual signed pulls reported.
Network implementation cross-check	The live-gate rule should survive more than one network environment.	PRyMordial passes the transported abundance row and AlterBBN resolves the closed-form rate response at high precision.

Table 10: Layered controls used to test the reaction and abundance construction.

The robustness content of the result is therefore not only numerical closeness. It is the fact that the same construction survives the sequence of source, reaction, coordinate, abundance, live-gate, one admitted PRyMordial abundance row, and one independent AlterBBN rate-response marker without converting the lithium result into an adjustable depletion parameter.

12.1 Falsifiability Conditions

The construction is falsifiable in several direct ways. It fails if the mass-seven agreement requires replacing $\ln \pi$ by a fitted optical depth, if the live ${}^7\text{Li}$ and ${}^7\text{Be}$ source terms

do not reproduce the $1/\pi$ survival relation in admitted BBN backgrounds, if the transported BBN coordinate spoils D/H or Y_p , or if the lithium agreement appears only after double-counting a diagnostic ${}^7\text{Be}$ output.

It also fails as a geometric reaction theory if the source-state construction cannot be carried into a coherent amplitude, if the $S_{12}(0)$ comparison breaks under the stated contraction, or if another reaction channel is asserted without a separate source-state construction. The nearest independent test is reconstruction of $\mathcal{A}_{C6}$ and I_{C7} from the archived support map. The second test is whether the closed-form K_{S0} branch remains compatible with independent $D(p, \gamma){}^3\text{He}$ measurements. The third is whether the fixed $\tau_7 = \ln \pi$ live-gate rule survives source-network implementations without changing the D/H and Y_p controls. Failure at any one of these layers narrows or rejects the combined claim. These failure conditions are useful because they make the paper testable by researchers who do not adopt the framework in advance.

Failure condition	Consequence for the claim
$\mathcal{A}_{C6}$ or I_{C7} cannot be reconstructed from the stated source-state rules	The nuclear source-state claim fails.
The $S_{12}(0)$ comparison moves outside the admitted band under the closed-form normalization	The commissioned reaction comparison or normalization fails.
The transported BBN coordinate fails D/H or Y_p before lithium is judged	The abundance background is inadmissible.
The live $\tau_7 = \ln \pi$ gate fails in another source-network implementation under the same abundance coordinate	The mass-seven transport claim is weakened or rejected.
Successful lithium requires changing τ_7 , η_{10}^{BBN} , or the S -factor normalization after seeing the result	The no-retuning claim fails.

Table 11: Concrete falsification ledger for the source, reaction, abundance-coordinate, and live-gate layers.

13 PRyMordial Admission Row and AlterBBN Rate-Response Marker

The closed-form $D(p, \gamma){}^3\text{He}$ normalization was propagated through the PRyMordial transported abundance row and an AlterBBN rate-response marker. The operative rate branch is fixed by $K_{S0} = \alpha\pi/(33 - 1/(7\pi)) = 0.0006956649414530029 \text{ eV b}$, $S_{12}^{\text{CPTG}}(0) = 0.2145003969219307 \text{ eV b}$, and $S_{12}^{\text{CPTG}}(6.64 \text{ keV}) = 0.2516322073981925 \text{ eV b}$. The PRyMordial transported row gives the admitted abundance comparison used in Table 9. The AlterBBN row is not used as an independent full abundance-admission row in this manuscript; it is a high-precision marker showing that the same closed-form $D(p, \gamma){}^3\text{He}$ rate normalization is resolved in a second network implementation. This distinction matters because the raw AlterBBN D/H marker is a rate-response diagnostic, whereas the admitted abundance row is the PRyMordial transported row.

For the PRyMordial transported row, the closed-form rate gives

$$\left(\frac{{}^7\text{Li}}{H}\right)_{\text{raw,PRy}} = 5.2668261732457650 \times 10^{-10} \quad (81)$$

and therefore

$$\left(\frac{{}^7\text{Li}}{H}\right)_{\text{gated,PRy}} = 1.6764828397556692 \times 10^{-10}. \quad (82)$$

For the AlterBBN high-precision rate-response marker, the closed-form rate gives

$$\left(\frac{{}^7\text{Li}}{H}\right)_{\text{raw,Alter}} = 5.4661777402483630 \times 10^{-10} \quad (83)$$

and the corresponding fixed-gate comparison value is

$$\left(\frac{{}^7\text{Li}}{H}\right)_{\text{gated,Alter}} = 1.7399384143588267 \times 10^{-10}. \quad (84)$$

Network	Raw ${}^7\text{Li}/H$	Gated ${}^7\text{Li}/H$	Lithium pull	Controls/result
PRyMordial	$5.2668261732457650 \times 10^{-10}$	$1.6764828397556692 \times 10^{-10}$	$+0.90593\sigma$	D/H and Y_p pass
AlterBBN	$5.4661777402483630 \times 10^{-10}$	$1.7399384143588267 \times 10^{-10}$	$+1.15975\sigma$	high-precision rate response resolved

Table 12: PRyMordial admission row and AlterBBN rate-response marker using the closed-form CPTG $D(p, \gamma){}^3\text{He}$ normalization. The PRyMordial row is the admitted transported abundance row. The AlterBBN row is not an independent full abundance-admission row in this manuscript; it is a high-precision marker showing that the same closed-form rate normalization is resolved in a second network implementation.

The network implementation cross-check is not interpreted as a free retuning of the baryon coordinate or as an empirical lithium-depletion factor. It is a source-network test of the live ${}^7\text{Li}$ and ${}^7\text{Be}$ transport equation at fixed optical depth, with the $D(p, \gamma){}^3\text{He}$ reaction branch normalized by the closed-form CPTG expression. The PRyMordial row supports the lithium comparison band inferred from old metal-poor stars at the admitted abundance-comparison level, while the AlterBBN row verifies the same rate branch as a separate high-precision implementation marker.

14 Relation to Existing Solution Classes

The result should be compared with the main classes of explanation normally used for the primordial lithium problem and for nuclear-rate sensitivity in BBN. The relevant benchmark is not whether a model can reduce lithium alone. The benchmark is whether one framework can connect a stated nuclear source-state construction to a coherent $D(p, \gamma){}^3\text{He}$ reaction amplitude, preserve the deuterium and helium BBN controls, reproduce the raw mass-seven excess, and then recover the lithium abundance through a fixed live $A = 7$ transport rule. Many existing lithium-problem solution classes are formulated primarily at the abundance, stellar-depletion, expansion-history, or nonstandard-particle level; the comparison here is the more specific source-to-reaction-to-abundance performance vector.

This combined claim is unusually restrictive for an alternative-gravity framework. Many modified-gravity or beyond-standard-model approaches enter BBN through an altered

expansion rate, an effective gravitational coupling, extra radiation, particle decays, entropy injection, or a shifted freeze-out history. Those mechanisms can change abundance predictions, but they normally leave the nuclear reaction source structure external to the gravity theory. CPTG instead makes the nuclear reaction chain and the mass-seven abundance transport part of the same geometric framework.

The comparison target is therefore not a single lithium number. It is a simultaneous admitted vector of nuclear and abundance observables,

$$\mathcal{C}_{\text{admit}} = \left\{ S_{12}(0), \frac{D}{H}, Y_p, \left(\frac{{}^7\text{Li}}{H} \right)_{\text{PRyMordial}} \right\}. \quad (85)$$

The independent AlterBBN result is kept as a marker vector rather than an admitted abundance vector,

$$\mathcal{M}_{\text{AlterBBN}} = \left\{ R_{pD}^{\text{CPTG}}(T), \left(\frac{{}^7\text{Li}}{H} \right)_{\text{AlterBBN}}, \tau_7 = \ln \pi \right\}. \quad (86)$$

The relevant comparison is whether one fixed geometric construction can keep the admitted row inside its stated observational controls while also surviving a separate implementation marker, without introducing a lithium-specific depletion coefficient. This is why the low-residual pattern in Tables 9 and 12 is important: the theory is not merely moving lithium into range, but doing so while retaining the surrounding nuclear and abundance checks with the actual signed pulls reported row by row.

The first comparison is standard BBN without a mass-seven transport gate. In that setting, deuterium and helium are successful controls, but the raw surviving mass-seven abundance remains high. CPTG keeps the successful background controls and changes only the live mass-seven transport sector. The raw abundance still displays the lithium problem before the gate is applied, so the calculation does not hide the discrepancy by changing the comparison target.

The second comparison is nuclear-rate retuning. A rate-retuning approach tries to alter one or more reaction rates enough to reduce the final mass-seven abundance. The risk is that a rate large enough to solve lithium can become difficult to reconcile with laboratory nuclear constraints or with the surrounding abundance network. In the present theory, the dimensional $D(p, \gamma){}^3\text{He}$ comparison is not used as a lithium knob. The zero-energy S -factor scale is closed form, the external comparison row is -0.149960σ , and the mass-seven reduction is carried by the separate live transport gate.

The third comparison is stellar depletion. Stellar depletion mechanisms attempt to explain why old metal-poor stars show less lithium at their surfaces than raw standard BBN predicts. CPTG does support the lithium abundance inferred from those old metal-poor stars, but it does not do so by assigning a star-by-star surface-depletion coefficient. The comparison target is the lithium abundance band inferred from old metal-poor stars; the mechanism is a primordial source-network transport rule acting on ${}^7\text{Li}$ and ${}^7\text{Be}$.

The fourth comparison is nonstandard particle or radiation physics. Such models may introduce decays, new species, entropy production, altered expansion, or additional ra-

diation degrees of freedom. CPTG instead uses a geometric transport law with a fixed optical depth,

$$\tau_7 = \ln \pi. \quad (87)$$

The corresponding survival factor is therefore

$$\exp(-\tau_7) = \frac{1}{\pi}. \quad (88)$$

This makes the lithium correction a fixed geometric map rather than a free abundance-sector parameter.

The fifth comparison is a purely phenomenological abundance rescaling. A post-processing rescaling could always be chosen to move lithium into an observed band. That is not the rule used here. The CPTG prescription is a live source-network equation for both mass-seven channels,

$$\frac{dY_7}{dt} \longrightarrow \left(\frac{dY_7}{dt} \right)_{\text{BBN}} - \Gamma_{7,\text{trans}} Y_7, \quad (89)$$

with the integrated optical depth fixed before the observational comparison. This is the difference between a fitted abundance multiplier and a geometric transport theory.

Approach	Main mechanism	CPTG distinction
Standard BBN	Successful D/H and helium, raw lithium excess remains	Retains D/H and helium while adding fixed live mass-seven transport
Nuclear-rate retuning	Modify reaction rates to reduce mass seven	Keeps the commissioned $S_{12}(0)$ comparison small-residual and separates the lithium gate from rate fitting
Stellar depletion	Reduce observed stellar-surface lithium after BBN	Supports the lithium band inferred from old metal-poor stars through primordial live transport, not an external stellar-surface coefficient
Modified expansion theories	Alter expansion, freeze-out, effective coupling, or radiation content	CPTG adds a reaction-specific geometric source chain and a fixed live mass-seven transport gate, rather than only modifying the background expansion history
Nonstandard particles or radiation	Add new species, decays, entropy, or expansion effects	Uses a fixed geometric optical depth $\ln \pi$ and survival factor $1/\pi$
Phenomenological rescaling	Multiply lithium by a chosen factor after the calculation	Applies a live ${}^7\text{Li}$ and ${}^7\text{Be}$ source-network term before final comparison

Table 13: Conceptual comparison between CPTG and common lithium-problem solution classes.

Requirement	Meaning
Reaction source construction	A theory must specify a nuclear source-state path, not only a final abundance multiplier.
Coherent amplitude	The source construction must produce a reaction amplitude or equivalent reaction-strength object.
S-factor contact	The theory must make contact with $D(p, \gamma)^3\text{He}$ through an astrophysical S-factor or equivalent rate interface.
D/H and Y_p preservation	The light-element background must remain admitted before lithium is judged.
Live mass-seven treatment	The mechanism must act on live ^7Li and ^7Be source-network variables, not only on a final displayed abundance.
Fixed lithium rule	The lithium reduction must not be an adjustable depletion coefficient.
Admitted row plus marker	The rule should survive one admitted abundance row plus a separate implementation marker.

Table 14: Minimum requirements for an external explanation to match the combined source-to-reaction-to-abundance performance vector tested here; the network requirement is framed as one admitted abundance row plus an independent implementation marker.

The practical comparison standard is therefore broader than the lithium endpoint. To make the same kind of claim, an external model would need to supply a reaction-specific nuclear source construction, preserve the deuterium and helium controls, account for the lithium band inferred from old metal-poor stars, and do so without converting the lithium correction into an adjustable abundance multiplier.

15 What the Construction Addresses and Why It Matters

The construction addresses a coupled problem rather than a single numerical discrepancy. First, it supplies a geometric source construction for a specific nuclear reaction. Second, it carries that source into the conventional nuclear-astrophysics language of an astrophysical S -factor and a thermonuclear rate. Third, it places the reaction in the transported BBN abundance coordinate. Fourth, it applies a fixed live transport law to the surviving mass-seven channels. Fifth, it compares the resulting abundances with observed comparison values using signed pulls.

The implication is that a geometric theory can be tested inside nuclear astrophysics without discarding the normal observables of the field. The theory still has to meet the S -factor comparison, the D/H control, the helium control, and the lithium observation. It is not enough to match one number. The same framework must keep several distinct comparison rows small at the same time.

The strongest practical implication is that the cosmological lithium problem can be reframed as a surviving mass-seven transport problem rather than only as a nuclear cross-section problem or a stellar-surface depletion problem. In this formulation, the raw BBN abundance is allowed to show the usual lithium excess. The geometric transport sector then acts on the live mass-seven channels with a fixed optical depth. Because the survival factor is $1/\pi$, the predicted final abundance is tightly constrained.

The broader implication is that nuclear reactions may admit geometric source organiza-

tion when the calculation is built from state support, current/source coefficients, coherent amplitude, and comparison-layer observables. The scope remains reaction-specific. The present paper establishes a working pattern for one important channel: source state, coherent amplitude, $S(E)$, thermonuclear rate, and abundance pulls. That pattern is the research value of the paper. It gives other researchers a compact set of equations to test, challenge, or extend.

16 Applications and Use Cases

The completed reaction-chain construction is useful because it gives nuclear astrophysics a concrete geometric object to test. It does not ask a BBN code to hide the lithium discrepancy by adjusting a final abundance. Instead, it supplies a closed-form $D(p, \gamma)^3\text{He}$ rate branch, a live $A = 7$ transport law, and an abundance-coordinate comparison that can be inserted into ordinary source-network calculations.

16.1 Baryonic Early-Universe Modeling

In standard cosmological modeling, unresolved abundance tensions are often explored by adding nonstandard matter sectors, nonstandard radiation content, altered decay histories, or other beyond-standard ingredients. The CPTG nuclear construction provides a different use case: it tests whether the light-element sector can be modeled through ordinary baryonic reaction networks plus a geometric transport law. In that role, the framework is useful as a baryonic early-universe modeling reference. It lets a researcher ask how far the primordial abundance problem can be carried without introducing an additional lithium-specific particle species or an adjustable abundance-depletion coefficient.

The direct application is to BBN scenario building. A modeler can run deuterium, helium, helium-three, and mass-seven comparisons using the fixed transported BBN coordinate, the closed-form $D(p, \gamma)^3\text{He}$ rate branch, and the locked $\tau_7 = \ln \pi$ mass-seven transport rule. The impact is a cleaner separation between geometric transport effects and speculative nonstandard-particle explanations.

16.2 BBN Software and Source-Network Upgrades

The framework can be implemented as a source-network upgrade to BBN codes. The $D(p, \gamma)^3\text{He}$ branch is inserted as

$$R_{pD}^{\text{CPTG}}(T) = \lambda_{pD} R_{pD}^{\text{table}}(T), \quad \lambda_{pD} = 1.0000018504518913, \quad (90)$$

with the closed-form dimensional normalization

$$K_{S0} = \frac{\alpha\pi}{33 - \frac{1}{7\pi}} = 0.0006956649414530029 \text{ eV b.} \quad (91)$$

In this implementation formula, $R_{pD}^{\text{table}}(T)$ is the unmodified carrier rate used by the named network implementation, and λ_{pD} is the fixed ratio between the closed-form CPTG $S_{12}(0)$ branch and that carrier normalization. It is not refitted inside the BBN abundance calculation.

The live mass-seven transport terms are inserted as

$$\frac{dY_{7\text{Li}}}{dt} \rightarrow \frac{dY_{7\text{Li}}}{dt} - \Gamma_{7,\text{trans}}(T)Y_{7\text{Li}} \quad (92)$$

and

$$\frac{dY_{7\text{Be}}}{dt} \rightarrow \frac{dY_{7\text{Be}}}{dt} - \Gamma_{7,\text{trans}}(T)Y_{7\text{Be}}, \quad (93)$$

with

$$\int \Gamma_{7,\text{trans}}(t) dt = \ln \pi. \quad (94)$$

This gives computational physicists a direct implementation recipe for P_{RY}Mordial-style, AlterBBN-style, or PRIMAT-style networks [2, 3, 4]. The practical impact is that the lithium-sector correction is represented as a live source-network term with a fixed optical depth, not as a hand-adjusted post-processing multiplier.

16.3 Laboratory Cross-Checks for Low-Energy Capture Measurements

The closed-form branch supplies an independent theoretical comparison value for the zero-energy astrophysical S -factor of the ${}^2\text{H}(p, \gamma){}^3\text{He}$ reaction:

$$S_{12}^{\text{CPTG}}(0) = 0.2145003969219307 \text{ eV b}. \quad (95)$$

This is useful for low-energy nuclear laboratories because direct measurements must be extrapolated through detector response, electron-screening controls, background modeling, and energy-window corrections. CPTG does not supersede a laboratory measurement; it provides a geometry-derived value that can be used as an external theoretical reference when comparing extrapolated $S_{12}(0)$ values.

The impact is diagnostic. If future precision measurements drift systematically away from the closed-form value, the discrepancy becomes a concrete test of the geometric reaction branch. If they converge toward it, the branch becomes a stronger independent cross-check on the low-energy extrapolation.

16.4 First-Generation Star and Primordial Lithium Modeling

Old, metal-poor stars provide the observational lithium band used in the comparison. Conventional explanations often invoke stellar-surface depletion, diffusion, turbulence, rotation, or other stellar-evolution effects to connect the observed stellar abundance to

a larger primordial value. The CPTG construction is useful because it separates the primordial mass-seven transport question from star-by-star surface depletion modeling.

In this framework, the source network may produce the usual high raw mass-seven abundance, while the live geometric transport rule maps the surviving ${}^7\text{Li}$ and ${}^7\text{Be}$ channels into a fixed primordial survival coordinate. Stellar-evolution models can then be tested against a lower primordial mass-seven reference coordinate rather than being forced to absorb the full standard-BBN lithium excess by themselves.

The impact is a cleaner division of labor: BBN and geometric transport set the primordial mass-seven coordinate, while stellar astrophysics can focus on residual surface and population effects.

16.5 Blueprint for Additional Geometric Nuclear Reactions

The present paper is reaction-specific. It does not claim that every nuclear channel is already solved. Its broader value is that it establishes a working architecture:

$$\text{source state} \rightarrow \text{coherent amplitude} \rightarrow S(E) \rightarrow N_A \langle \sigma v \rangle \rightarrow \text{abundance pulls.} \quad (96)$$

This architecture can be cloned and tested on other channels, including channels that affect helium-four, beryllium-seven flow, or other reaction bottlenecks in primordial and stellar environments. Each extension must supply its own source-state construction, current/source coefficients, amplitude rule, dimensional comparison, and network validation.

The impact is programmatic. The paper gives a reproducible template for a possible field of geometric nuclear physics: not a claim that all subatomic reactions are already determined by geometry, but a concrete method for testing whether specific nuclear reaction strengths and abundance-network effects can be derived from geometric source organization.

17 Claim Level and Limitations

The commissioned reaction channel establishes a native source-to-amplitude construction for $D(p, \gamma){}^3\text{He}$, a closed-form dimensional $S_{12}(0)$ normalization, a thermonuclear-rate interface, and a transported BBN abundance test. The operative result uses the closed-form $K_{S0} = \alpha\pi/(33 - 1/(7\pi))$ normalization throughout.

The live mass-seven gate establishes a fixed source-network transport rule for ${}^7\text{Li}$ and ${}^7\text{Be}$. The optical depth is $\ln \pi$, and the resulting survival factor is $1/\pi$. This does support the lithium abundance inferred from old metal-poor stars as the observational comparison target, with the lower-bound caveat attached to that stellar inference. What is not claimed here is an external stellar-depletion fit, a star-by-star stellar-evolution model, a solution of every abundance systematic, or a universal solution of all nuclear reaction channels. Extension to other reactions requires a separate source-state construction and separate comparison tests.

The BBN comparison is a controlled source-network and coordinate-layer test. It is not a full global cosmological parameter fit. Future work should test broader nuclear networks, alternative observational abundance priors, additional source-network implementations, and reaction channels beyond $D(p, \gamma)^3\text{He}$.

Item	Established in this paper	Not claimed
Reaction channel	Source-to-amplitude construction for $D(p, \gamma)^3\text{He}$	Universal solution of all nuclear reactions
S -factor	Closed-form $S_{12}(0)$ normalization, external cross-check row, and high-precision network response	Universal dimensional scale for all reaction channels
BBN coordinate	Transported BBN abundance coordinate with D/H and Y_p controls	Free retuning of the acoustic baryon coordinate
Mass-seven gate	Live ^7Li and ^7Be source-network transport with $\tau_7 = \ln \pi$; supports the lithium band inferred from old metal-poor stars	External stellar-depletion fit or star-by-star stellar-evolution model
Network implementation cross-check	PRyMordial locked-gate row and AlterBBN high-precision rate-response marker with displayed fixed-gate lithium diagnostic	Exhaustive survey of all BBN codes and priors

Table 15: Claim-level summary for the reaction and abundance-transport calculation.

18 Reproducibility Recipe and Data Ledger

A researcher testing the formulas independently can follow the sequence below. First, reconstruct $\mathcal{A}_{C6}$ from the retained support-index registry, W_{ik} source-weight map, current/source coefficient tuple, and scalar source components supplied in the companion reproducibility archive. The printed manuscript defines the contraction, but the finite 68×5 support map is a data object and must be checked before the dimensional S -factor layer is used. Second, verify $I_{C7} = |\mathcal{A}_{C6}|^2$. Third, use the transported BBN coordinate

$$\eta_{10}^{\text{BBN}} = 5.998071834744. \quad (97)$$

Fourth, compute or load the raw BBN source-network abundances at that coordinate. Fifth, keep deuterium and helium as background controls. Sixth, evolve the live ^7Li and ^7Be channels with the transport terms

$$-\Gamma_{7,\text{trans}} Y_{7\text{Li}} \quad (98)$$

and

$$-\Gamma_{7,\text{trans}} Y_{7\text{Be}}. \quad (99)$$

Seventh, enforce

$$\int \Gamma_{7,\text{trans}}(t) dt = \ln \pi. \quad (100)$$

Eighth, compare the final mass-seven abundance to the lithium comparison band using

$$\Delta_{7\text{Li}} = \frac{(^7\text{Li}/H)_{\text{CPTG}} - (^7\text{Li}/H)_{\text{obs}}}{\sigma_{7\text{Li}}}. \quad (101)$$

All abundance results should be reported with the raw mass-seven value, the gated mass-seven value, the D/H pull, the helium pull, and the lithium pull. The companion CPTG repository is the public reproducibility resource for the supporting code packages, benchmark implementations, support-map archives, and research references [12]. The companion report archive listed in Table 16 supplies the detailed hash ledgers rather than requiring the manuscript to inline every support-table row. The principal companion archive is:

CPTG_DPGammaHe3_GeometricNuclearReactionTheoryFrozenFullReport_v10_671_r01.zip

The network-output archive used for the final source-network tables is:

CPTG_DPGammaHe3_GeometricNuclearReactionTheoryFrozenFullReport_v10_671_r01.zip/
EVIDENCE_OUTPUT_ZIPS/CPTG_v10_670_outputs_r01.zip

Reproduction object	Archive locator	Use in this manuscript
Report archive	CPTG.DPGammaHe3_GeometricNuclearReactionTheoryFrozenFullReport_v10.671_r01.zip	Starting point for source-to-amplitude, rate, BBN, and source-network closure replay
Manifest ledger	LEDGERS/FREEZE_PACKAGE_SHA256_MANIFEST.csv	Verifies archive contents before using any extracted evidence object
Formula and parameter ledgers	LEDGERS/FORMULA_LEDGER.csv LEDGERS/PARAMETER_LEDGER.csv	Source for equations, constants, row-362 values, and comparison quantities
Evidence/source audit ledgers	LEDGERS/EVIDENCE_PACKAGE_AUDIT.csv LEDGERS/SOURCE_PACKAGE_AUDIT.csv	Reopens the included output ZIPS and source packages without relying on external context
Network-output archive	CPTG.DPGammaHe3_GeometricNuclearReactionTheoryFrozenFullReport_v10.671_r01.zip/ EVIDENCE_OUTPUT_ZIPS/ CPTG_v10_670_outputs_r01.zip and internal OUTPUTS/	Reproduces the PRyMordial/AlterBBN closure values used for the final tables
Report entry point	REPORTS/CPTG.Geometric_Nuclear_Reaction_Theory_Frozen_Full_Report_v10.671_r01.md	Human-readable reconstruction path and audit notes for the source-to-rate result

Table 16: Reproducibility data ledger for reconstructing the source map, contraction, rate insertion, and BBN output parsing. Full archive paths and exact file hashes are retained in the listed ledgers.

19 Conclusion

This paper presents a geometric nuclear-reaction theory for $D(p, \gamma)^3\text{He}$ as an extension of the CPTG lithium solution. The construction links four layers that are usually treated separately: a nuclear source-state construction, a coherent reaction amplitude, an S -factor and rate interface, and a live source-network abundance gate. The same framework preserves deuterium and helium controls while moving raw mass-seven output into the lithium abundance band inferred from old metal-poor stars.

The commissioned reaction channel reaches the nuclear comparison layer through

$$\mathcal{A}_{C6} = 17.5595746398194 \quad (102)$$

and

$$I_{C7} = 308.3386615313886. \quad (103)$$

With the closed-form dimensional scale, the S -factor comparison gives

$$\Delta_S = -0.149960\sigma. \quad (104)$$

The abundance layer is then tested at the transported BBN coordinate rather than by direct retuning of the acoustic baryon coordinate. The raw mass-seven abundance remains high before the gate, so the lithium problem is not hidden by the coordinate choice. The locked live transport rule instead acts on ${}^7\text{Li}$ and ${}^7\text{Be}$ with

$$\tau_7 = \ln \pi \quad (105)$$

and

$$Y_{7,\text{CPTG}} = \frac{Y_{7,\text{raw}}}{\pi}. \quad (106)$$

The source-network comparisons using the closed-form rate scale give

$$\Delta_{7\text{Li},\text{PRyMordial}} = +0.90593\sigma \quad (107)$$

and

$$\Delta_{7\text{Li},\text{AlterBBN}} = +1.15975\sigma. \quad (108)$$

The retained PRyMordial abundance projection gives a gated lithium pull of $+0.90593\sigma$ while retaining the D/H and Y_p controls, and the AlterBBN marker row resolves the same closed-form rate response at ppm precision. These rows show a low-residual pattern across reaction and abundance layers, rather than a lithium-only endpoint adjustment.

The result is best described as a reaction-specific geometric nuclear-reaction result with a fixed abundance-transport extension. It gives a concrete target for independent reproduction or falsification: reconstruct the source-state contraction, carry it to the S -factor and rate layer, evaluate the transported BBN coordinate, apply the live $A = 7$ gate with fixed $\ln \pi$ optical depth, and report all abundance pulls. If any of those layers fails, the combined claim fails.

The scope remains bounded. The paper does not present a completed theory of all nuclear reactions, and each additional reaction channel requires its own source-state construction and comparison tests. What is established here is that the commissioned $D(p, \gamma){}^3\text{He}$ channel and the locked mass-seven transport rule can be written as one explicit source-to-reaction-to-abundance chain, with small signed pulls in the stated comparison rows. That is the central research value of the construction.

Future work should extend the source-state construction to additional reactions, test the live transport law in additional network implementations, examine broader abundance priors, and determine whether other nuclear channels admit comparably compact geometric source organization. The present paper supplies the formulas and comparison protocol needed for those tests.

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